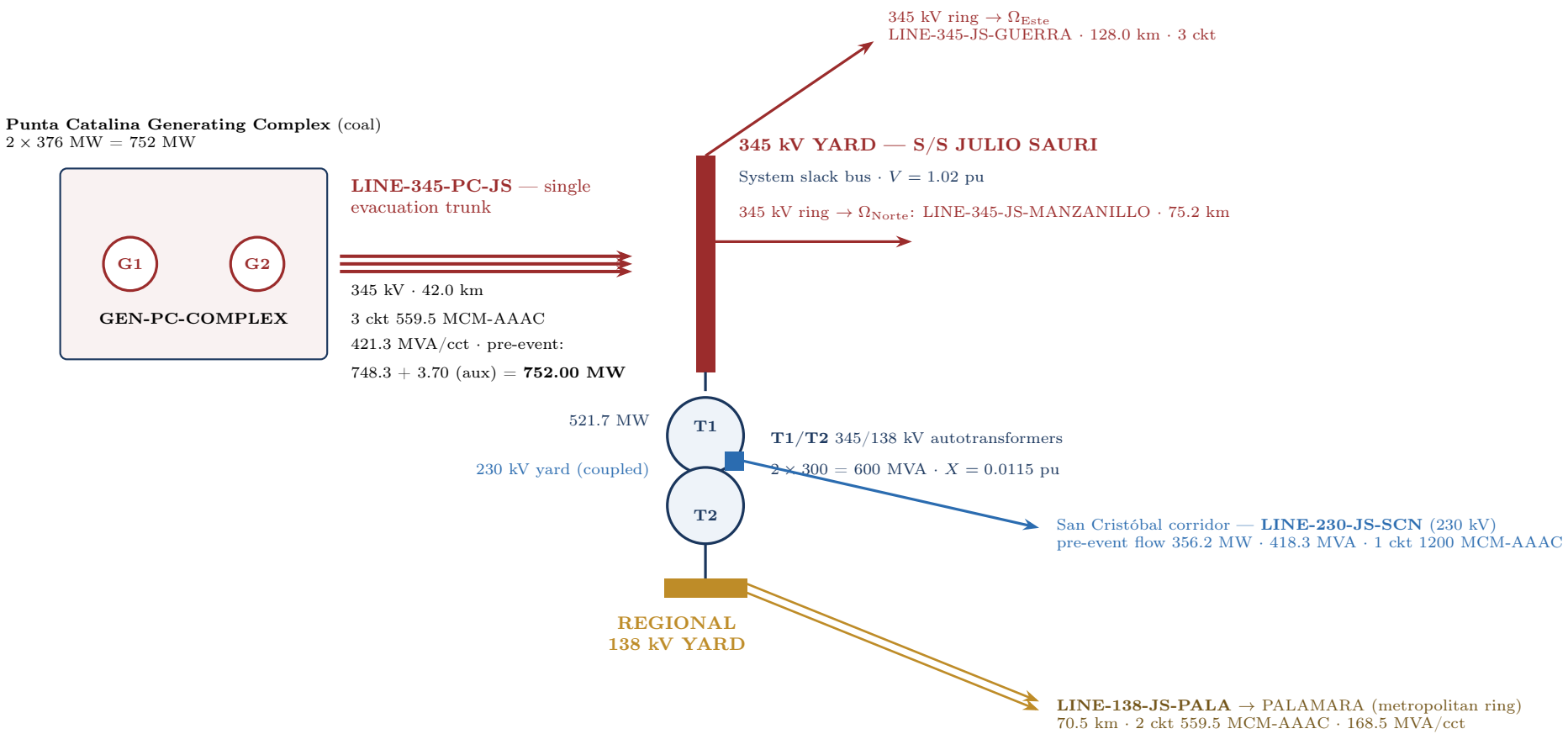
Kirchhoff Decoupling at Julio Sauri Substation

Single 345 kV Evacuation Trunk **LINE-345-PC-JS** and 345/138 kV Transformation — SENI-142 benchmark



Canonical reserve-sizing event (common mode: LINE-345-PC-JS)

Trunk loss isolates 752.00 MW → redispatch: 687.40 MW (ΔP_g) + 58.50 MW (storage) + 6.10 MW (slack) = 752.00 MW (error 0.0000)

Required reserve: $R_t^{\text{req}} = \max\big(\max_g P_g^{\max},\, P_{\text{PC}}^{\text{cmplx}}\big) = \max(376,\, 752) = 752$ MW $\quad\Rightarrow\quad$ 345 and 138 kV yards decoupled through $X_{T1/T2} = 0.0115$ pu